



Regulatory Complaint Investigation and CAPA Report

Confidential - AI Draft

Document No.	DP-M4-20260707-THEDOCPL
Classification	Confidential - AI Draft
Generated	2026-07-07T15:47:18Z
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Firmware	v3.4
Framework	ISO 13485
Prepared By	DocPlus+ M4 Evidence and Documentation Layer

This document was generated from live M1 graph facts and M2 agent outputs. Every summarized claim is tied to captured agent state, evidence citations, or audit findings.

Table of Contents

Section	Title
1	Document Profile and Controlled Metadata
2	Regulatory Format Principles - ALCOA+ Framework
3	Executive Summary
4	Complaint Investigation Record
5	Requirements Traceability Matrix
6	Design History File Index
7	EU MDR Technical Documentation - Annex II / III
8	Complaint Investigation Workflow and Execution Trace
9	Root Cause Analysis
10	Evidence and Provenance
11	CAPA Report
12	Cybersecurity / SBOM Narrative
13	Electronic Records - 21 CFR Part 11
14	AuditShadow and Trace Decay
15	Cross-Standard Rendering Matrix
16	Document Control and Approval

This output follows a formal regulatory-document pattern: controlled metadata first, then complaint facts, investigation logic, objective evidence, risk decision, CAPA plan, review gates, and approval placeholders.

1. Document Profile and Controlled Metadata

Document Title	DocPlus+ Regulatory Complaint Investigation and CAPA Report
Document Number	DP-M4-20260707-THEDOCPL
Revision	A - AI Generated Draft
Complaint ID	CPL-20260707-0B12
CAPA ID	CAPA-20260707-7EC3
RTM ID	RTM-DP-0B127EC3
Classification	Confidential - AI Draft for quality and regulatory review
Generated Timestamp	2026-07-07T15:47:18Z
System of Record	DocPlus+ M4 Evidence and Documentation Layer
Device	DocPlus+ Pulse Oximeter (DEV-PULSE-OX)
Current Firmware	v3.4
Affected Component	display_task
Regulatory Framework	ISO 13485

Data Quality and Source Provenance

This report references 49 graph fact(s): 12 controlled, 22 extracted, 6 inferred, 6 synthetic, 3 internal.

Uncontrolled graph facts appear in this report and are visibly tagged. [DEMO DATA - NOT CONTROLLED EVIDENCE] means the item cannot support CAPA closure without controlled verification. [INFERRED DATA - REVIEW REQUIRED] means the item was inferred and requires reviewer confirmation before final use.

Source Type	Referenced Fact Count
controlled	12
extracted	22
inferred	6
synthetic	6
internal	3

Digital-Twin Prototype Disclaimer

Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Document Purpose

This report documents the complaint intake, AI-assisted investigation record, retrieved objective evidence, risk decision, CAPA recommendations, audit-style gaps, and corrective action controls. It is intended as a controlled draft that a Quality or Regulatory reviewer can verify, approve, or return for correction.

2. Regulatory Format Controls

ALCOA+ Data Integrity Mapping

Principle	DocPlus+ Implementation
Attributable	Prepared by DocPlus+ M4; generated timestamp and device scope are recorded.
Legible	A4 PDF with selectable text, structured headings, repeatable tables, and page numbers.
Contemporaneous	Generated at the time of complaint-report request from live agent state.
Original	Marked as AI Draft until Quality and Regulatory reviewers approve.
Accurate	Risk, CAPA, evidence, audit, and trace sections are populated from structured state.
Complete	Includes intake, investigation, evidence, risk, CAPA, audit, trace, and approval fields.
Consistent	Uses the same DocPlus+ controlled layout for all complaint-report exports.
Enduring	Saved as a PDF artifact under the application output directory.
Available	Downloadable through the M4 document endpoint and traceable by document number.

Applicable Regulatory Basis

Framework	Relevance to This Report
ISO 13485	Complaint handling, CAPA discipline, documented information, traceable quality records.
21 CFR Part 820	Design controls, complaint files, CAPA, and objective evidence expectations.
EU MDR	Technical documentation, post-market surveillance, vigilance, and risk-benefit continuity.
21 CFR Part 11	Electronic record controls, auditability, attribution, and durable retrieval.

3. Executive Summary

Severity High	Risk Critical / RPN 20	Reportable True
Evidence Confidence medium (score 0.83)	Evidence Class Mix Candidate evidence - requires Quality review: 1, Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 3	Evidence Items 14
Audit Findings 2	Trace Decay Alerts 2	

Complaint: The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings disappear, and the alarm does not trigger until the device is restarted.

Agent conclusion: Complaint for DEV-PULSE-OX was classified as High severity affecting display_task. Current deterministic RPN is 20.

Risk Evidence Confidence Basis

Weighted evidence-class score 0.83 from 14 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=3. CAPA-blocking evidence-class item count: 4. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Confidence drivers: RUN-TC-DISP-015-V3-4: controlled_verification, RUN-TC-DISP-013-V3-4: controlled_verification, RUN-TC-DISP-017-V3-4: controlled_verification, RUN-TC-DISP-018-V3-4: controlled_verification, RUN-TC-DISP-012-V3-4: controlled_verification, RUN-TC-DISP-011-V3-4: controlled_verification, RUN-TC-DISP-020-V3-4: controlled_verification, RUN-TC-DISP-014-V3-4: contro...

Quality Review Decision

The complaint should remain open until objective evidence is reconciled, CAPA ownership is assigned, and any major or critical audit findings affecting the complaint scope are remediated or justified.

4. Complaint Investigation Record

Field	Generated Value	Regulatory Basis
Complaint ID	CPL-20260707-0B12	ISO 13485 8.2.2
Date Received	2026-07-07T15:47:18Z	21 CFR Part 11
Device UDI	UDI-DI-08717648-825630	EU MDR Article 27 / 21 CFR 830
Model / Version	DOCPLUS+-PULSE-OXIMETER / v3.4	IEC 62304 Section 8
Complaint Source	Post-market complaint intake	ISO 13485 8.2.2
Complaint Narrative	The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings disappear, and the alarm does not trigger until the device is restarted.	21 CFR 820 complaint file
Patient Impact	Potential delayed awareness of a safety or status condition.	EU MDR Article 87
MDR Reportable?	Yes	EU MDR Article 87 / 21 CFR 803
CAPA Initiated?	Yes - CAPA-20260707-7EC3	21 CFR 820.100
Affected Component	display_task	ISO 13485 8.5.2
Closure Status	Pending Engineering Approval - Simulated Evidence Sufficient: Formal Engineering review must approve the simulated findings and decide whether additional physical verification is required.	ISO 13485 8.2.2

Reportability Decision Logic

Condition	DocPlus+ Assessment	Action
Death or serious injury	Not stated in complaint text.	Monitor for escalation during intake and investigation.
Potential harm or near miss	Potential delayed awareness of a safety or status condition.	Treat as reportability review required.
Device relationship	Complaint maps to Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys....	Keep CAPA open until engineering evidence confirms or rules out relationship.
Malfunction/use issue	Reported condition indicates possible device malfunction or use-related failure.	Verify recurrence risk and effectiveness of controls.

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
CMP-004	6	Pulse rate display froze until the sensor was reconnected. [DEMO DATA - NOT CONTROLLED EVIDENCE]	synthetic [DEMO DATA - NOT CONTROLLED EVIDENCE]
CPL-100-030	5	Pulse waveform is not displayed. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-051	5	High SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-052	5	Low SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-053	5	High pulse rate alarm missing. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

5. Requirements Traceability Matrix

Req ID	Requirement	Scope / Source	Relevance	Evidence	Test Case	Status / Result
REQ-DISP-015	Screen freezes during operation.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 10.	verified_current - Controlled current verification	TC-DISP-015	current and usable - Pass
REQ-SPO2-005	The display shall present SpO2 and pulse rate readings without ambiguity. [DEMO DATA - NOT CONTROLLED EVIDENCE]	strong - M1 requirement scope [DEMO DATA - NOT CONTROLLED EVIDENCE]	Matched complaint terms against Design Control; relevance score 8.	missing - Candidate evidence - requires Quality review [DEMO DATA - NOT CONTROLLED EVIDENCE]	TEST-005, TEST-025	usable with caution - Pass
REQ-DISP-011	OLED/LCD screen remains blank.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-011	current and usable - Pass
REQ-DISP-012	Display flickers intermittently.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-012	current and usable - Pass
REQ-DISP-013	Display brightness cannot be adjusted.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-013	current and usable - Pass
REQ-DISP-014	Dead pixels appear on the screen.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-014	current and usable - Pass
REQ-DISP-016	Display shows garbled characters.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-016	current and usable - Pass
REQ-DISP-017	Display orientation is incorrect.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-017	current and usable - Pass
REQ-DISP-018	Touch buttons do not respond.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-018	current and usable - Pass
REQ-DISP-019	Physical navigation buttons stick.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-019	current and usable - Pass
REQ-DISP-020	Display backlight turns off unexpectedly.	strong - RTM_M aster.xlsx	Matched complaint terms against display_task; relevance score 7.	verified_current - Controlled current verification	TC-DISP-020	current and usable - Pass
REQ-MEAS-030	Pulse waveform is not displayed.	strong - RTM_M aster.xlsx	Matched complaint terms against measurement_core; relevance score 5.	verified_current - Controlled current verification	TC-MEAS-030	current and usable - Pass

Trace Decay Risk Statement

Rows marked STALE are not filler rows: they are generated from the live Trace Decay result and indicate verification links that require review before the device can claim current-firmware evidence.

6. Design History File Index

#	Element	Document ID	Generated Unique Content	Status
1	Planning	DDP-DP-0B127EC3	Design plan for complaint-impact review	Open
2	Design Input	URS-DP-0B127EC3	Complaint-linked requirement review for The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings d...	Draft
3	Design Output	SDD-DP-0B127EC3	Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys... design output references mapped	Draft
4	Design Review	DRM-DP-0B127EC3	Cross-functional review required	Pending
5	Verification	VPR-DP-0B127EC3	Objective evidence required for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys... on v3.4	Open
6	Validation	VAL-DP-0B127EC3	Use-condition validation required for The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings d...	Open
7	Design Transfer	DTR-DP-0B127EC3	Release readiness blocked until evidence reconciled	Pending
8	Design Changes	DCR-DP-0B127EC3	Change/service impact assessment linked to Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys...	Open
9	DHF Index	DHFI-DP-0B127EC3	DocPlus+ index for DocPlus+ Pulse Oximeter	Generated

Validation Gap Check

The validation section remains open because the complaint describes a user-reported condition involving Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys.... DocPlus+ therefore does not mark the DHF complete until validation, verification, or justified non-reproducibility evidence is linked.

7. EU MDR Technical Documentation - Annex II / III

Annex Ref	Section	DocPlus+ Generated Content
Annex II Section 1	Device Description	UDI UDI-DI-08717648-825630 scoped to complaint involving Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys....
Annex II Section 3	Design and Manufacturing Information	Design/change impact for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys... is carried into design output review.
Annex II Section 4	GSPR Compliance	Reported condition maps to safety/performance review: The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings disappear, and the al....
Annex II Section 5	Benefit-Risk Analysis	Critical risk complaint requires residual-risk review before closure.
Annex II Section 6.1	Verification and Validation	Objective verification evidence must be reconciled against the complaint scope.
Annex III Section 1.3	PMS / PSUR Input	Complaint is retained as a post-market signal for trend monitoring.
Annex III Section 1.5	Vigilance Reporting	Reportability remains open until human regulatory review confirms decision.

Team NB Structure Check

The package preserves a cohesive structure with section titles, metadata, and cross-references so that information is locatable by a reviewer rather than buried in a generic AI narrative.

Record Boundary

The report is generated only from live application state: M1 graph context, M2 agent outputs, M4 evidence retrieval, AuditShadow findings, and Trace Decay alerts. It does not certify regulatory closure until the signature block is completed by authorized reviewers.

8. Complaint Investigation Workflow and Execution Trace

8.1 Structured Complaint Intake Detail

Device ID	DEV-PULSE-OX
Affected Component	display_task (COMP-DISPLAY-UI)
Severity	High
Timeline	post-change or delayed event
Symptom Codes	SYMPTOM_DISPLAY_FREEZE, SYMPTOM_MISSING, SYMPTOM_ALARM_DELAY, SYMPTOM_DOCPLUS, SYMPTOM_PULSE, SYMPTOM_OXIMETER, SYMPTOM_DISPLAY, SYMPTOM_FREEZES
Match Terms	display, freeze, freezes
Similar Prior Complaints	5

Complaint Narrative

The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings disappear, and the alarm does not trigger until the device is restarted.

Related Prior Complaints

Complaint ID	Similarity	Summary	Source
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CPL-100-030	5	Pulse waveform is not displayed. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-051	5	High SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-052	5	Low SpO2 alarm does not trigger. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]
CPL-100-053	5	High pulse rate alarm missing. [INFERRED DATA - REVIEW REQUIRED]	inferred [INFERRED DATA - REVIEW REQUIRED]

8.2 Investigation Chronology

Time	Agent	Action	Status	Detail
2026-07-07T15:43:18.441244Z	langgraph_supervisor	route to complaint_intake	info	Recorded
2026-07-07T15:43:18.441244Z	complaint_intake	extract structured complaint	started	Recorded
2026-07-07T15:43:39.416807Z	complaint_intake	extract structured complaint	completed	Elapsed 20981.7 ms
2026-07-07T15:43:39.416807Z	langgraph_supervisor	handoff check: similar incidents preserved for root cause	info	Structured event metadata recorded
2026-07-07T15:43:39.433405Z	langgraph_supervisor	route to root_cause	info	Recorded
2026-07-07T15:43:39.433405Z	root_cause	generate complaint-specific failure hypotheses	started	Recorded
2026-07-07T15:43:45.518957Z	root_cause	generate complaint-specific failure hypotheses	completed	Elapsed 6086.1 ms
2026-07-07T15:43:45.548505Z	langgraph_supervisor	route to firmware_traceability_ripple_check	info	Recorded
2026-07-07T15:43:45.548505Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	started	Recorded
2026-07-07T15:43:45.550118Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-07T15:43:46.430417Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-07T15:43:46.430417Z	firmware_traceability_ripple_check	hypothesis capability check	info	Structured event metadata recorded
2026-07-07T15:43:46.430417Z	firmware_traceability_ripple_check	run firmware traceability ripple checks	completed	Elapsed 883.3 ms
2026-07-07T15:43:46.458315Z	langgraph_supervisor	route to evidence	info	Recorded
2026-07-07T15:43:46.458315Z	evidence	retrieve and validate evidence	started	Recorded
2026-07-07T15:43:47.718721Z	evidence	component-scoped evidence retrieval	info	Structured event metadata recorded
2026-07-07T15:43:47.718721Z	evidence	retrieve and validate evidence	completed	Elapsed 1261.9 ms
2026-07-07T15:43:47.749170Z	langgraph_supervisor	route to risk	info	Recorded

Chronology Interpretation

The sequence confirms the complaint passed through intake, root-cause hypothesis generation, firmware traceability ripple checking, evidence retrieval, risk scoring, CAPA drafting, and final reasoning. This preserves a reviewer-readable process trail for the AI investigation rather than presenting the CAPA as an unsupported narrative.

Trace AI - Pipeline Observability

Trace AI is operational engineering telemetry for this report generation. It records pipeline execution, LLM metadata, and retrieval/tool metadata only. It is not clinical evidence, not regulatory objective evidence, and not the same as Trace Decay or the Firmware Traceability Ripple Check.

Total wall-clock time	239716.21 ms
Agent steps	10
LLM calls	5
Tool/retrieval calls	25
Tokens	15871 total (12218 prompt / 3653 completion)
Longest agent	openai_reasoning (174149.88 ms)

Agent Step Trace

Step ID	Agent	Action	Status	Elapsed	Sub-events
step-58f2699399	complaint_intake	extract structured complaint	success	20981.73 ms	LLM 1; tools 4
step-a685482a70	root_cause	generate complaint-specific failure hypotheses	success	6086.06 ms	LLM 1; tools 2
step-e10f933afd	firmware_traceability_ripple_...	run firmware traceability ripple checks	success	883.25 ms	LLM 0; tools 2
step-e59b4d9136	evidence	retrieve and validate evidence	success	1261.89 ms	LLM 0; tools 3
step-38d8a557d7	risk	calculate ISO 14971 risk	success	71.75 ms	LLM 0; tools 2
step-51bd871598	capa	draft CAPA with citations	success	979.44 ms	LLM 0; tools 4
step-0a03eb2bc1	openai_reasoning	final complaint reasoning	success	153585.88 ms	LLM 2; tools 0
step-f7932fda65	audit_shadow	run adversarial audit	success	23280.17 ms	LLM 0; tools 5
step-e8bd206230	openai_reasoning	final audit reasoning	success	20564.0 ms	LLM 1; tools 0
step-9dc227ab45	cybersecurity	scan SBOM components against NVD	success	12022.04 ms	LLM 0; tools 3

LLM Call Metadata

Call ID	Agent	Task	Model	Status	Tokens	Latency
llm-2cd78c623a	complaint_intake	symptom_classification	gpt-5	success	1883 (1361/522)	6185.87 ms
llm-824d31c067	root_cause	root_cause_hypotheses	gpt-4o-mini	success	5993 (5326/667)	6018.41 ms
llm-b5a910ca4c	openai_reasoning	complaint_final_reasoning	gpt-5	error	not returned	77000.82 ms
llm-55aff4787b	openai_reasoning	complaint_final_reasoning	gpt-5	error	not returned	76581.87 ms
llm-32ba75b0a5	openai_reasoning	audit_final_reasoning	gpt-5	success	7995 (5531/2464)	20562.8 ms

Tool and Retrieval Call Metadata

Call ID	Agent	Tool	Query / Target	Results	Latency
tool-d91f128526	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		22.87 ms
tool-08b0b43e00	complaint_intake	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		23.51 ms
tool-7015613d68	complaint_intake	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	11678.63 ms
tool-a1849dd4ab	complaint_intake	graph_tools.find_similar_complaints	term=The DocPlus+ pulse oximeter display freezes after Bluetooth connec...	5	28.43 ms
tool-f647944b75	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-DISPLAY-UI	398	42.87 ms
tool-9c42974dd3	root_cause	graph_tools.get_graph_neighborhood	node_id=COMP-DISPLAY-UI	12	22.77 ms
tool-b706e9df33	firmware_traceability_rip...	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		23.32 ms
tool-7f5b251a96	firmware_traceability_rip...	graph_tools.run_ripple	device_id=DEV-PULSE-OX; new_version=v3.5; changed_components=['COMP-DIS...	10	878.47 ms
tool-40f134e1f1	evidence	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		27.68 ms
tool-c895874d32	evidence	graph_tools.get_component_scoped_ev...	device_id=DEV-PULSE-OX; component_id=COMP-DISPLAY-UI; component_name=di...	12	110.87 ms
tool-24055b7c9f	evidence	vector_tools.retrieve_documents_debug	device_id=DEV-PULSE-OX; query=The DocPlus+ pulse oximeter display freez...	8	1149.16 ms
tool-740ab9626e	risk	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		21.63 ms
tool-8ca2962f5f	risk	graph_tools.open_capa_count	device_id=DEV-PULSE-OX	2	71.18 ms
tool-cf262e78cd	capa	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		23.35 ms
tool-201fadb2d3	capa	graph_tools.get_audit_scope	device_id=DEV-PULSE-OX	120	283.32 ms
tool-eb67aab521	capa	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		22.51 ms
tool-418c355a41	capa	graph_tools.get_capa_context	device_id=DEV-PULSE-OX	1	689.1 ms
tool-e42b29d2a6	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		25.0 ms
tool-5d51099aab	audit_shadow	graph_tools.resolve_device_id	device_id=DEV-PULSE-OX		22.15 ms
tool-c2e6fe9433	audit_shadow	graph_tools.get_device_context	device_id=DEV-PULSE-OX	15	11775.06 ms

5 additional tool/retrieval call(s) are retained in structured Trace AI data.

9. Root Cause Analysis

ID	Hypothesis	Probability	Affected Area	Key Why Chain
HYP-001	display_task: Bluetooth Connectivity Issue [DEMO DATA - NOT CONTROLLED EVIDENCE]	33%	COMP-DISPLAY-UI	Device connects to Bluetooth. Display task is initiated. Simultaneously, the display freezes due to a bug. [DEMO DATA - NOT CONTROLLED EVIDENCE]
HYP-002	display_task: Software/Firmware Bug [INFERRED DATA - REVIEW REQUIRED]	39%	COMP-DISPLAY-UI	Firmware processes display data. Bluetooth connection triggers an erroneous state. Display task fails to update, causing freeze. [INFERRED DATA - REVIEW REQUIRED]
HYP-003	display_task: Alarm Functionality Failure [INFERRED DATA - REVIEW REQUIRED]	28%	COMP-DISPLAY-UI	Alarm system initialized. Freezing display prevents alarm from being depicted or triggered properly. [INFERRED DATA - REVIEW REQUIRED]

HYP-001 Evidence Balance

Evidence For	The complaint specified that the display freezes after Bluetooth connection. Prior incidents show connectivity-related issues causing delays and display problems. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Evidence Against / Gaps	No direct evidence ruling out Bluetooth-related software bugs. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Similar Incident Cross-Check	CMP-004 - Pulse rate display froze until the sensor was reconnected. Pattern is consistent with connectivity issues. [DEMO DATA - NOT CONTROLLED EVIDENCE]
Probability Rationale	Given the direct relation of the symptom to the Bluetooth connection, the probability is relatively high. [DEMO DATA - NOT CONTROLLED EVIDENCE]

HYP-002 Evidence Balance

Evidence For	Existed issues with software-induced display problems, such as similar freezing behaviors. Existing evidence points to potential firmware bugs. [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No comprehensive tests confirming firmware behavior under Bluetooth conditions. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CPL-100-030 - High SpO2 alarm not triggering due to a software issue. Pattern consistent with that of software bugs causing display issues. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	The documentation suggests issues in firmware could lead to failures like freezing, giving a higher probability on this hypothesis. [INFERRED DATA - REVIEW REQUIRED]

HYP-003 Evidence Balance

Evidence For	Complaints allege alarm issues linked to display problems. Historical incidents show alarm functionalities affected by display freezing. [INFERRED DATA - REVIEW REQUIRED]
Evidence Against / Gaps	No direct evidence specifically linking the alarm to display freezing in the current case. [INFERRED DATA - REVIEW REQUIRED]
Similar Incident Cross-Check	CPL-100-051 - High SpO2 alarm not triggering. This aligns with the observed alarm failures in the complaint. [INFERRED DATA - REVIEW REQUIRED]
Probability Rationale	The lack of triggering alarms displayed in historical incidents lends validity to this hypothesis, but its confirmation would depend on further investigation. [INFERRED DATA - REVIEW REQUIRED]

Root Cause Rationale

1. The display of the DocPlus+ pulse oximeter freezes when connected via Bluetooth, potentially due to software bugs in the connectivity stack.
2. The workflow collected 8 evidence item(s) and 3 hypothesis item(s).
3. Open CAPA, stale evidence, similar incidents, and readiness gaps are the main graph-grounded drivers.

Primary Root Cause Position

For this generated draft, the leading position is that the complaint is associated with Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys... and the evidence/control set tied to the reported condition. The position remains provisional until engineering review confirms or rules out the suspected cause with objective evidence.

10. Evidence and Provenance

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-DISP-015 / TC-DISP-015 / RUN -TC-DISP-015-V3-4	Controlled current verification	Yes	96%	REQ-DISP-015: Screen freezes during operation. Test TC-DISP-015: Verification test TC-DISP-015; result=Pass; firmware=v3.4. Evidence RUN-TC-DISP-015-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-DIS P-015/TC-DISP-015, Confidence: 0.96]
REQ-DISP-013 / TC-DISP-013 / RUN -TC-DISP-013-V3-4	Controlled current verification	Yes	96%	REQ-DISP-013: Display brightness cannot be adjusted. Test TC-DISP-013: Verification test TC-DISP-013; result=Pass; firmware=v3.4. Evidence RUN-TC-DISP-013-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW...	[Source: REQ-DIS P-013/TC-DISP-013, Confidence: 0.96]
REQ-DISP-017 / TC-DISP-017 / RUN -TC-DISP-017-V3-4	Controlled current verification	Yes	96%	REQ-DISP-017: Display orientation is incorrect. Test TC-DISP-017: Verification test TC-DISP-017; result=Pass; firmware=v3.4. Evidence RUN-TC-DISP-017-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-DIS P-017/TC-DISP-017, Confidence: 0.96]
REQ-DISP-018 / TC-DISP-018 / RUN -TC-DISP-018-V3-4	Controlled current verification	Yes	96%	REQ-DISP-018: Touch buttons do not respond. Test TC-DISP-018: Verification test TC-DISP-018; result=Pass; firmware=v3.4. Evidence RUN-TC-DISP-018-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Ver...	[Source: REQ-DIS P-018/TC-DISP-018, Confidence: 0.96]
REQ-DISP-012 / TC-DISP-012 / RUN -TC-DISP-012-V3-4	Controlled current verification	Yes	96%	REQ-DISP-012: Display flickers intermittently. Test TC-DISP-012: Verification test TC-DISP-012; result=Pass; firmware=v3.4. Evidence RUN-TC-DISP-012-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4...	[Source: REQ-DIS P-012/TC-DISP-012, Confidence: 0.96]

Source	Evidence Class	Support	Conf.	Excerpt	Citation
REQ-DISP-011 / TC-DISP-011 / RUN-TC-DISP-011-V3-4	Controlled current verification	Yes	96%	REQ-DISP-011: OLED/LCD screen remains blank. Test TC-DISP-011: Verification test TC-DISP-011; result=Pass; firmware=v3.4. Evidence RUN-TC-DISP-011-V3-4: result=Pass; firmware=v3.4. EVID-TEST-REPORT-COMPLAINT100-REGRESSION-FW-V3-4: TEST REPORT Complaint100 Regression FW-v3.4 Ve...	[Source: REQ-DISP-011/TC-DISP-011, Confidence: 0.96]

10.1 Firmware Traceability Ripple Check

This section reports a firmware-traceability check: for each root-cause hypothesis, it examines whether linked verification test records match the complaint or candidate firmware version, or are stale/mismatched. This is a graph-based traceability signal, not a behavioral model or physical test of the device. It identifies where verification evidence may be out of date, not what the device would actually do under the reported failure condition.

Hypothesis	Analysis Type	Result	Interpretation	Evidence Class
HYP-002	Firmware traceability ripple check	fail	HYP-002 evaluates display_task: Software/Firmware Bug. The firmware traceability ripple check found 10 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verification coverage for this hypothesis, no...	Firmware traceability signal - informational, not controlled verification evidence
HYP-001	Firmware traceability ripple check	fail	HYP-001 evaluates display_task: Bluetooth Connectivity Issue. The firmware traceability ripple check found 10 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verification coverage for this hypothe...	Firmware traceability signal - informational, not controlled verification evidence
HYP-003	Firmware traceability ripple check	fail	HYP-003 evaluates display_task: Alarm Functionality Failure. The firmware traceability ripple check found 10 linked verification test record(s) that do not match candidate firmware v3.5; this flags stale or mismatched verification coverage for this hypothes...	Firmware traceability signal - informational, not controlled verification evidence

11. CAPA Report

Product	DocPlus+ Pulse Oximeter
Software Version	v3.4
CAPA Status	Pending Engineering Approval - Simulated Evidence Sufficient
Priority	High
Risk Classification	Major - potential impact to Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys...
Evidence Confidence	medium (score 0.83)
Evidence Class Mix	Candidate evidence - requires Quality review: 1, Controlled current verification: 10, Firmware traceability signal - informational, not controlled verification evidence: 3

Risk Evidence Confidence Basis

Weighted evidence-class score 0.83 from 14 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=3. CAPA-blocking evidence-class item count: 4. Rule: Risk evidence confidence is the average of per-item evidence-class weights (controlled_verification=1.00, historical_controlled=0.70, simulated=0.45, candidate=0.25, no_evidence=0.00). The final label is capped below high when any CAPA-closure-blocking evidence class is present, because those items cannot support final closure until controlled verification is attached.

Engineering Approval Gate

This CAPA is supported by 10 controlled verification record(s) and 3 digital-twin or firmware-traceability simulation result(s) consistent with the investigation. The scoped closure chain contains no synthetic source nodes, candidate evidence, stale historical evidence, or no-evidence markers. Formal Engineering review must approve the simulated findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Problem Statement

A high priority complaint was opened for DocPlus+ Pulse Oximeter. The user reported: The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings disappear, and the alarm does not trigger until the device is restarted. In practical terms, the concern is that a user may not receive a safety or status notification at the expected time. The complaint record confirms the reported condition; the exact technical cause and final correction remain under investigation until objective evidence is reviewed.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Complaint Summary

Reported condition: The DocPlus+ pulse oximeter display freezes after Bluetooth connection. SpO2 readings disappear, and the alarm does not trigger until the device is restarted. Affected area: Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability. Reported timing or trigger: After a reported change or preceding event. Observed symptom signals: inconsistent or inaccurate measurement readings. No patient or user injury is stated in the complaint text; the risk review must still confirm potential safety impact.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Immediate Containment Actions

Preserve the complaint evidence for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Investigation

Relevant device requirements, verification records, risk controls, and open quality actions were reviewed for the complaint scope. The investigation focused on the reported condition and the affected area: Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability. For this complaint the scoped area is Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability, with emphasis on alarm or notification timing, event logs, and risk-control behavior. Engineering and Quality reviewed the available design expectations, test records, risk controls, event evidence, and related quality actions. 2 verification records need Quality review; 1 related quality action remains open. The scoped verification evidence is available for Quality review. Risk evidence basis: Weighted evidence-class score 0.83 from 14 item(s): Candidate evidence - requires Quality review=1, Controlled current verification=10, Firmware traceability signal - informational, not controlled verification evidence=3. CAPA-blocking evidence-class item count: 4.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Root Cause Analysis

The current root-cause hypothesis is: The display of the DocPlus+ pulse oximeter freezes when connected via Bluetooth, potentially due to software bugs in the connectivity stack. This is a probable cause, not a final confirmed cause, because the CAPA record still needs engineering reproduction results, configuration or software-version review, logs, and approved verification evidence showing whether the suspected cause is truly responsible.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Why CAPA Remains Open

The CAPA is not blocked for lack of evidence. The scoped evidence chain has controlled verification plus digital-twin or firmware-traceability simulation support, so the remaining gate is formal Engineering approval. Engineering must review the simulated findings against notification timing and user-visible response, approve or reject their applicability, and record whether physical/controlled verification is still required. Risk evidence confidence is medium (score 0.83); Medium uncertainty: 10 of 14 evidence item(s) are current controlled verification, but 4 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure. This CAPA is supported by 10 controlled verification record(s) and 3 digital-twin or firmware-traceability simulation result(s) consistent with the investigation. The scoped closure chain contains no synthetic source nodes, candidate evidence, stale historical evidence, or no-evidence markers. It remains pending formal Engineering approval before closure; simulated results support the investigation but do not substitute for physical or controlled verification. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Risk Assessment

The main hazard is that a user may not receive a safety or status notification at the expected time. The current risk level is Critical and the complaint should remain in active risk review under ISO 13485 until patient impact, recurrence potential, and residual risk acceptability are documented. Risk evidence confidence is medium (score 0.83); Medium uncertainty: 10 of 14 evidence item(s) are current controlled verification, but 4 item(s) remain historical, simulated, candidate, or no-evidence class and cannot support CAPA closure.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Corrective Actions

Reproduce or simulate the reported condition for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability under the reported use conditions. Software/System Engineering should review alarm or notification timing, event logs, and risk-control behavior, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Engineering must review and approve the simulated findings before closure; if Engineering rejects the simulation or deems it insufficient, open targeted physical or controlled verification before final approval.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Preventive Actions

Update the design, service, or release checklist so future changes affecting Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability include complaint-specific verification for inconsistent or inaccurate measurement readings. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Verification of Effectiveness

Effectiveness is supported by controlled verification records plus simulated analysis of notification timing and user-visible response. Before closure, an Engineering reviewer must approve the digital-twin or firmware-traceability simulation result, document why it is sufficient for this complaint, or require additional physical verification.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Effectiveness Criteria

The CAPA is effective when the behavior for notification timing and user-visible response is supported by current controlled records and the digital-twin or firmware-traceability simulation is formally approved by Engineering. Closure requires documented Engineering sign-off; additional physical verification is required only if Engineering does not accept the simulated support.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Lessons Learned

Future complaint handling for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability should connect user-reported symptoms, objective evidence, risk controls, and verification records early. Similar changes or service decisions should include test coverage for the reported condition and review of whether existing controls remain effective.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

CAPA Closure

Closure decision: pending Engineering approval. This CAPA is supported by 10 controlled verification record(s) and 3 digital-twin or firmware-traceability simulation result(s) consistent with the investigation. The scoped closure chain contains no synthetic source nodes, candidate evidence, stale historical evidence, or no-evidence markers. Formal Engineering review must approve the simulated findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Evidence class gate: Controlled current verification, Firmware traceability signal - informational, not controlled verification evidence

Technical evidence identifiers, requirement IDs, audit findings, and source citations are retained in the evidence and audit appendices. They are not repeated here so the CAPA remains readable for Quality, Engineering, Regulatory, and non-technical reviewers.

11.1 CAPA Action Plan

CAPA Record	CAPA-20260707-7EC3
Complaint Reference	CPL-20260707-0B12
CAPA Type	Corrective and preventive action
Current Status	Pending Engineering Approval - Simulated Evidence Sufficient
Engineering Approval Gate	This CAPA is supported by 10 controlled verification record(s) and 3 digital-twin or firmware-traceability simulation result(s) consistent with the investigation. The scoped closure chain contains no synthetic source nodes, candidate evidence, stale historical evidence, or no-evidence markers. Formal Engineering review must approve the simulated findings and decide whether additional physical verification is required. Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.

Action Plan

Purpose	Owner	Required Action	How Closure Is Proven
Immediate measures	Quality / Support	Preserve the complaint evidence for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability, including returned-unit information, event logs, service records, photos, user statements, configuration, lot or serial traceability, and software or hardware version where applicable. Screen recent complaints for similar symptoms and issue interim support, service, or use instructions if the risk review shows potential user or patient impact.	Evidence is preserved and simulated findings are available for Engineering review.
Fix the issue	Software/System Engineering	Reproduce or simulate the reported condition for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability under the reported use conditions. Software/System Engineering should review alarm or notification timing, event logs, and risk-control behavior, identify the confirmed failure mode, correct the issue or document why it cannot be reproduced. Engineering must review and approve the simulated findings before closure; if Engineering rejects the simulation or deems it insufficient, open targeted physical or controlled verification before final approval.	Engineering approves the digital-twin/simulated finding or records that physical verification is required.
Prevent recurrence	Engineering / Quality	Update the design, service, or release checklist so future changes affecting Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, System Reliability include complaint-specific verification for inconsistent or inaccurate measurement readings. Add regression or trend monitoring where appropriate, and require evidence review before Quality approves a similar change or field disposition.	Future changes include Engineering review of any simulated evidence before approval.
Check effectiveness	Quality Assurance	Effectiveness is supported by controlled verification records plus simulated analysis of notification timing and user-visible response. Before closure, an Engineering reviewer must approve the digital-twin or firmware-traceability simulation result, document why it is sufficient for this complaint, or require additional physical verification.	Closure is proven by Engineering sign-off on the simulated support plus Quality/Regulatory approval.

Acceptance Criteria for Closure

Closure Gate	Required Evidence
Reported condition	The reported condition is supported by controlled records and simulated investigation findings.
Affected function	Notification timing and user-visible response is covered by the scoped controlled and simulated evidence chain.
Engineering review	Engineering formally approves the digital-twin or firmware-traceability simulation, or requires physical verification.
Prototype disclaimer	Hackathon prototype disclaimer: this AI-generated investigation is supported by digital-twin or firmware-traceability simulation, not real hardware verification. Simulated results may support investigation triage, but they require formal human engineering approval before any real-world quality, regulatory, release, or closure reliance.
Approval	Engineering, Quality, and Regulatory reviewers approve the completed CAPA record before closure.

12. Cybersecurity / SBOM Narrative

This section is device-level cybersecurity surveillance. It is not complaint evidence and it does not change the Risk Agent score automatically. It records what the Cybersecurity Agent found by reading the SBOM and querying the National Vulnerability Database for the CPE candidates in the workbook.

SBOM components	35
Components queried	25
NVD findings	7
Query date	2026-07-04
Source type	extracted
Cache status	cache_hit

Executive Severity Rollup

Critical	High	Medium	Low	None
0	2	5	0	0

VEX-style exploitability status is shown per CVE. The current automated SBOM/NVD match does not include device-specific exploitability determinations, so findings default to Under Investigation until Engineering documents whether the vulnerable code path is affected, fixed, not affected, or not reachable in this device configuration.

SBOM Component Coverage

Component	Version	Supplier	CPE Candidate	NVD Result
Bootloader	1.8.0	Device OEM / Secure Boot Team	cpe:2.3:o:device_oem:pulseox_bootloader:1.8.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Embedded Firmware	2.4.3	Device OEM / Embedded Software Team	cpe:2.3:o:device_oem:pulseox_firmware:2.4.3:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Operating System / RTOS	10.5.1	Amazon Web Services, Inc.	cpe:2.3:o:amazon:freertos:10.5.1:****:*	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
Device Drivers	1.6.2	Device OEM / Driver Integration Team	cpe:2.3:a:device_oem:pulseox_device_drivers:1.6.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Signal Processing Software	3.2.1	Device OEM / Clinical Algorithm Supplier	cpe:2.3:a:device_oem:pulseox_signal_processing:3.2.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Measurement Control Logic	2.1.0	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_measurement_control:2.1.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Alarm Management Module	2.0.5	Device OEM / Safety Software Team	cpe:2.3:a:device_oem:pulseox_alarm_manager:2.0.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
User Interface Module	1.7.4	Device OEM / UI Software Team	cpe:2.3:a:device_oem:pulseox_ui_module:1.7.4:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Settings Configuration Manager	1.9.1	Device OEM / Embedded Software Team	cpe:2.3:a:device_oem:pulseox_settings_manager:1.9.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Data Storage Manager	1.5.0	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_data_storage_manager:1.5.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Patient Record Manager	1.3.2	Device OEM / Data Platform Team	cpe:2.3:a:device_oem:pulseox_patient_record_manager:1.3.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Communication Stack	2.2.0	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_communication_stack:2.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Bluetooth Stack	5.3.0	Nordic Semiconductor ASA	cpe:2.3:a:nordicsemi:bluetooth_low_energy_stack:5.3.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Wi-Fi Stack	2.10	w1.fi project / Device OEM Integration	cpe:2.3:a:w1.fi:wpa_supplicant:2.10:****:	open_cves_found: 1 CVE record(s) returned by NVD for this CPE.
TCP/IP Stack	2.1.3	lwIP project	cpe:2.3:a:lwip_project:lwip:2.1.3:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
USB Interface Software	1.4.1	Device OEM / Connectivity Team	cpe:2.3:a:device_oem:pulseox_usb_interface:1.4.1:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
File System	3.8.1	littlefs project / Device OEM Integration	cpe:2.3:a:littlefs_project:littlefs:3.8.1:****:***	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Logging Module	1.2.8	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_logging_module:1.2.8:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Error Handling Module	1.1.6	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_error_handling:1.1.6:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Time Management Module	1.0.9	Device OEM / Platform Software Team	cpe:2.3:a:device_oem:pulseox_time_manager:1.0.9:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Self-Test Module	1.3.5	Device OEM / Test Software Team	cpe:2.3:a:device_oem:pulseox_self_test:1.3.5:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Firmware Update Manager	2.0.2	Device OEM / Release Engineering Team	cpe:2.3:a:device_oem:pulseox_update_manager:2.0.2:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Authentication Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authentication_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.

Component	Version	Supplier	CPE Candidate	NVD Result
Authorization Module	1.2.0	Device OEM / Security Software Team	cpe:2.3:a:device_oem:pulseox_authorization_module:1.2.0:****:*	no_known_cves: No known CVEs found for this component/version combination as of 2026-07-04.
Cryptographic Library	3.0.13	OpenSSL Software Foundation	cpe:2.3:a:openssl:openssl:3.0.13:****:*	open_cves_found: 5 CVE record(s) returned by NVD for this CPE.

NVD CVE Findings - Structured Vulnerability Records

Vulnerability Record - CVE-2024-28115

CVE ID / NVD Reference	CVE-2024-28115 https://nvd.nist.gov/vuln/detail/CVE-2024-28115
CVSS Score / Severity / Vector	HIGH / CVSS 8.8 / v3.1 Vector: CVSS:3.1/ AV:L/ AC:L/ PR:L/ UI:N/ S:C/ C:H/ I:H/ A:H
CWE ID(s)	CWE-284: Improper Access Control, NVD-CWE-Other: Other weakness classification
Affected Component + Version	Operating System / RTOS 10.5.1
Vulnerability Description	NVD classifies the CVE under CWE-284, NVD-CWE-Other; review the public description for deployment-specific impact.
Exploitability / Exposure Context	Local or physically mediated exploitation should be assessed against service, update, or peripheral workflows. SBOM exposure: No direct exposure unless network stacks or external interfaces run as RTOS tasks.
Recommended Controls	review vendor fix status, confirm exploit preconditions, and document whether compensating controls cover the affected interface. SBOM controls: Minimal RTOS configuration, stack overflow checks, watchdog, task priority review, patch monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2024-28115

Vulnerability Record - CVE-2023-52160

CVE ID / NVD Reference	CVE-2023-52160 https://nvd.nist.gov/vuln/detail/CVE-2023-52160
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:R/ S:U/ C:H/ I:N/ A:N
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Wi-Fi Stack 2.10
Vulnerability Description	NVD describes an authentication or access-control bypass condition in the affected component.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Wi-Fi radio, WPA2/WPA3 negotiation, outbound cloud network.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: WPA2/WPA3, credential encryption, no ad-hoc mode, network allow-list, update monitoring.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2023-52160

Vulnerability Record - CVE-2009-1390

CVE ID / NVD Reference	CVE-2009-1390 https://nvd.nist.gov/vuln/detail/CVE-2009-1390
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-287: Improper Authentication
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes incomplete TLS certificate-chain validation, where a connection may be accepted because one certificate in an otherwise untrusted chain is trusted.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	verify full certificate-chain validation, require trusted root and hostname match, and regression-test rejected-chain cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-1390

Vulnerability Record - CVE-2009-3765

CVE ID / NVD Reference	CVE-2009-3765 https://nvd.nist.gov/vuln/detail/CVE-2009-3765
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3765

Vulnerability Record - CVE-2009-3766

CVE ID / NVD Reference	CVE-2009-3766 https://nvd.nist.gov/vuln/detail/CVE-2009-3766
CVSS Score / Severity / Vector	MEDIUM / CVSS 6.8 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:P/ I:P/ A:P
CWE ID(s)	CWE-310: Cryptographic Issues
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a certificate or TLS validation weakness in the affected secure-communication path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	update the TLS library, enforce strict certificate validation, and regression-test invalid certificate cases. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3766

Vulnerability Record - CVE-2009-3767

CVE ID / NVD Reference	CVE-2009-3767 https://nvd.nist.gov/vuln/detail/CVE-2009-3767
CVSS Score / Severity / Vector	MEDIUM / CVSS 4.3 / v2.0 Vector: AV:N/ AC:M/ Au:N/ C:N/ I:P/ A:N
CWE ID(s)	CWE-295: Improper Certificate Validation
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes certificate host-name validation that mishandles embedded NUL characters in the certificate Common Name, enabling a crafted certificate-name mismatch.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	reject embedded NUL characters in certificate names, prefer patched TLS hostname-validation routines, and test crafted certificate names. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2009-3767

Vulnerability Record - CVE-2019-0190

CVE ID / NVD Reference	CVE-2019-0190 https://nvd.nist.gov/vuln/detail/CVE-2019-0190
CVSS Score / Severity / Vector	HIGH / CVSS 7.5 / v3.1 Vector: CVSS:3.1/ AV:N/ AC:L/ PR:N/ UI:N/ S:U/ C:N/ I:N/ A:H
CWE ID(s)	NVD-CWE-noinfo: No CWE information
Affected Component + Version	Cryptographic Library 3.0.13
Vulnerability Description	NVD describes a denial-of-service condition that may let an attacker disrupt availability of the affected software path.
Exploitability / Exposure Context	Relevant when this component validates TLS peers, update endpoints, cloud links, or other certificate-backed channels. SBOM exposure: Used by outbound TLS, update signature verification, and local encryption features.
Recommended Controls	patch affected code paths, constrain exposed interfaces, and confirm watchdog/recovery behavior for availability loss. SBOM controls: Disable weak ciphers, certificate pinning, patch monitoring, crypto configuration review.
VEX-Style Status	Under Investigation
Status	Open/Patch Required. Requires review; NVD returned this CPE as affected and no local patch disposition is recorded in the SBOM.
Source / Reference	extracted; NVD CVE API 2.0 https://nvd.nist.gov/vuln/detail/CVE-2019-0190

13. Electronic Records - 21 CFR Part 11

Requirement	DocPlus+ Implementation	Part 11
System Validation	DocPlus+ generation path is deterministic and testable through FastAPI route and PDF render checks.	11.10(a)
Audit Trail	Generation record links CPL-20260707-0B12, CAPA-20260707-7EC3, timestamp, and agent trace events.	11.10(e)
Access Controls	AI output is marked draft and requires QE/RA approval before controlled use.	11.10(d)
Electronic Signatures	Signature block captures reviewer, approver, date, and meaning of signature.	11.50
Record Integrity	PDF is generated as a stable artifact with a deterministic document number and retrievable filename.	11.10(c)
Human-Readable Copy	Selectable-text PDF is available through the document download endpoint.	11.10(b)
Retention	Artifact is retained under output/pdf and can be indexed by M5 or a QMS repository.	11.10(c)

Generation Log Summary

Generation Hash	0B127EC3
Input Fingerprint	0B127EC3AA6258F8
AI Draft Status	Pending human quality and regulatory review
Low Confidence Handling	Evidence below 70% must be reviewed before release; current generated evidence remains visible with confidence values.

14. AuditShadow and Trace Decay

AuditShadow findings

Finding(s)	Requirement(s)	Risk	Observation	Remediation
FINDING-010	REQ-SPO2-005	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]
FINDING-011	REQ-SPO2-005	Major	Requirement is touched by 1 open CAPA record(s), so readiness cannot be treated as clean. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Close or justify open CAPA impact before audit claim. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Trace Decay alerts

Requirement(s)	Test(s)	Tested FW	Required FW	Reason
REQ-SPO2-005	TEST-005, TEST-025	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

14.1 Audit Data Interpretation and Corrective Action

AuditShadow and Trace Decay use the live M1 traceability graph as the controlled source for verification status. When the report flags missing evidence, it means the requirement does not have a formal VERIFIED_BY test link in that graph. It does not mean M4 found no supporting material. In this run, M4 retrieved 14 candidate evidence item(s), but candidate snippets remain draft support until a Quality reviewer links them to a requirement, test record, firmware version, and controlled source identifier.

Signal	What It Means	Source	Corrective Action
AuditShadow graph evidence gap	1 requirement finding(s) lack a VERIFIED_BY test link in M1.	M1 readiness score and traceability matrix	Create or link verification test evidence to each requirement and rescore readiness.
M4 retrieved evidence	14 candidate item(s) exist, but are not automatically controlled objective evidence.	M2 complaint pipeline and M4 evidence retrieval	Promote accepted snippets into controlled evidence nodes with requirement IDs, source, confidence, and firmware.
Trace Decay	2 firmware freshness alert(s) require review before release or approval.	M2 Trace Decay over M1 firmware/test links	Rerun verification on current firmware or document an approved equivalence rationale.
CAPA linkage	Open CAPA or complaint risk can keep readiness from being clean even when evidence exists.	M1 graph relationships and M2 risk assessment	Close, justify, or explicitly reference CAPA impact before final regulatory approval.

Reason and Corrective Action

The root issue is a traceability-linkage gap rather than a simple absence of all evidence. Corrective action is to synchronize M4 retrieved evidence back into the M1 graph as controlled verification evidence, then rerun AuditShadow and Trace Decay. Until that link exists, the report correctly treats the material as candidate support and keeps the audit finding open.

14.2 AuditShadow Detail Appendix

Finding(s)	Requirement(s)	Reference	Risk	Observation	Remediation
FINDING-010	REQ-SPO2-005	IEC 62366-1	Major	M1 graph has no VERIFIED_BY test evidence linked for this requirement. M4 may have retrieved candidate evidence, but it is not controlled objective verification evidence until Quality links it to the requirement, test record, source, and version/configuration metadata. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Link or create verification test evidence with current version/configuration metadata, then rerun AuditShadow. [DEMO DATA - NOT CONTROLLED EVIDENCE]
FINDING-011	REQ-SPO2-005	IEC 62366-1	Major	Requirement is touched by 1 open CAPA record(s), so readiness cannot be treated as clean. [DEMO DATA - NOT CONTROLLED EVIDENCE]	Close or justify open CAPA impact before audit claim. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Audit Interpretation

Major and critical findings should be triaged before report approval. A graph-linked evidence gap means the requirement is present in the M1 graph while still lacking a controlled verification test relationship acceptable for submission or inspection. Candidate RAG evidence must be reviewed, accepted, and linked before the finding can close.

14.3 Trace Decay Detail Appendix

Requirement(s)	Test(s)	Tested Firmware	Required Firmware	Reason
REQ-SPO2-005	TEST-005, TEST-025	v3.4	v3.4	Test is synthetic/demo backfill and cannot be treated as controlled verification evidence. [DEMO DATA - NOT CONTROLLED EVIDENCE]

Traceability Impact Statement

Trace Decay identifies verification evidence that may no longer represent the active or proposed firmware state. These alerts are not final nonconformities by themselves, but they are mandatory review inputs for release, CAPA, and audit-readiness decisions.

15. Cross-Standard Rendering Matrix

Element	FDA / QMSR	EU MDR	ISO 13485
Safety Requirement	Design input under QMSR; linked to verification evidence.	GSPR and Annex I safety/performance claim.	ISO 13485 Clause 7.3.3 design input.
Test Evidence	DHF verification record; current configuration/version required.	Annex II Section 6.1 V&V evidence.	Clause 7.3.6 design verification.
Complaint Record	Complaint file CPL-20260707-0B12 with malfunction assessment.	Post-market vigilance signal under Articles 83/87.	Complaint handling under Clause 8.2.2.
CAPA Record	CAPA-20260707-7EC3 opened for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys....	Trend/CAPA input for PMS and vigilance review.	Corrective action under Clause 8.5.2.
Risk Documentation	Risk analysis tied to design controls and complaint file.	Benefit-risk update in Annex II Section 5.	Risk management continuity through production/post-production.
Product Documentation	Lifecycle impact for Alarm Notification, Display and UI, Measurement and Sensor Performance, Connectivity and Data Transfer, Sys....	Design information and change-control evidence.	Validation and change-control evidence required.

Regulatory Lens Note

The same live graph and agent state are rendered under three regulatory lenses. This report therefore does not duplicate content blindly; it reformats the same evidence chain for FDA/QMSR, EU MDR, and ISO 13485 review contexts.

16. Document Control and Approval

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Generated from	Live M2 complaint pipeline, M1 graph context, M4 evidence retrieval, and device-level SBOM/NVD scan when attached
Generated at	2026-07-07T15:47:18Z
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Role	Name	Signature	Date
Quality Reviewer			
Regulatory Reviewer			